

GOVERNMENT OF INDIA
MINISTRY OF DEFENCE
Defence Research & Development Organisation (DRDO)
Centre for Airborne Systems (CABS), Bengaluru – 560 037



TECHNICAL REPORT
AIRBORNE SURVEILLANCE RADAR SYSTEM
MARK II – PERFORMANCE EVALUATION AND FIELD TRIAL ANALYSIS
Report No.: CABS/TR/2024/MK-II/001

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1. INTRODUCTION

The Airborne Surveillance Radar System (ASRS) Mark II is an advanced next-generation indigenous airborne Early Warning and Control System (EWACS) developed by the Centre for Airborne Systems (CABS), Bengaluru, under the aegis of the Defence Research & Development Organisation (DRDO). This report presents a comprehensive evaluation of the system performance subsequent to the completion of Phase III field trials conducted over the Rajasthan Test Range during December 2024 – January 2025.

The primary objective of the ASRS Mk II programme is to provide the Indian Air Force (IAF) with a superior indigenous surveillance capability featuring extended detection range and enhanced ECCM performance, thereby reducing dependence on imported systems and contributing to the national mission of Atmanirbhar Bharat in the domain of advanced defence electronics.

2. SCOPE OF THE REPORT

This technical report covers the following aspects of the ASRS Mk II system evaluation:

- Radar performance parameters against defined Key Performance Indicators (KPIs)
- System architecture and sub-system integration verification
- Reliability, Maintainability, and Availability (RMA) assessment
- Field trial data analysis over a 45-day operational window
- Recommendations for Mark III development and system upgradation

Additionally, this report documents the integration outcomes of the Advanced Electronic Warfare Suite (AEWS) which was incorporated into the Mk II configuration subsequent to recommendations raised during the Mk I evaluation. The AEWS integration was validated across all frequency bands of interest during the Phase III trials.

3. SYSTEM TECHNICAL SPECIFICATIONS

The following table presents the verified technical parameters of the ASRS Mk II system as measured during the Phase III field evaluation. All values have been independently verified by the DRDO Quality Assurance Directorate (QAD) and the Integrated Defence Staff (IDS) Technical Inspection Team.

Parameter	Specification	Achieved Value
Operating Frequency Band	L-Band (1–2 GHz)	1.35 GHz (Centre)
Detection Range (Air Target)	600 km (min.)	750 km
Altitude Coverage	100 m – 18,000 m	50 m – 18,500 m
Number of Simultaneous Tracks	250 (minimum)	300 tracks
Data Link Range	400 km	450 km
Radar Rotation Rate	6 RPM	6 RPM
Transmitter Peak Power	2.0 MW	2.05 MW
Pulse Width	10 μs / 50 μs	10 μs / 50 μs
MTBF	600 hours (min.)	650 hours
Operating Temperature	-20°C to +55°C	-25°C to +58°C
AEWS Integration	Full Band Coverage	Validated

4. PERFORMANCE METRICS & OBSERVATIONS



Figure 1: Radar Coverage Map – Detection Range 750 km (Phase III Trials)

4.1 Field Trial Performance Summary

Field trials conducted at the Rajasthan Test Range yielded data across 1,024 sorties over a 45-day evaluation window. The ASRS Mk II demonstrated performance exceeding the defined KPIs across all major parameters. Key observations are summarised as follows:

- False Alarm Rate (FAR) maintained below 10⁻⁶ per scan during all trial phases
- Detection probability (Pd) of 0.96 achieved against RCS = 1 m² targets at 680 km range

- Electronic Counter-Countermeasures (ECCM) validated against simulated jamming scenarios
- Identification Friend or Foe (IFF) Mode S integration confirmed operational
- System availability achieved at 98.1% over the 45-day trial window
- Data link demonstrated at 450 km range with end-to-end latency below 400 ms

The Enhanced Data Link (EDL) sub-system demonstrated consistently superior throughput at extended ranges owing to the incorporation of adaptive beamforming technology at the ground segment. Satellite relay integration trials confirmed seamless handover between line-of-sight and beyond-line-of-sight communication links.

4.2 Competitive Comparison Table

System Parameter	ASRS Mk II (DRDO)	Baseline Req.	Status
Detection Range	750 km	600 km	EXCEEDED
Track Capacity	300 tracks	250 tracks	EXCEEDED
MTBF	650 hrs	600 hrs	EXCEEDED
Jamming Resistance	83 dB/MHz	75 dB/MHz	EXCEEDED
Power Consumption	210 kW	220 kW (max)	MET
System Weight	3,680 kg	3,800 kg (max)	MET
AEWS Integration	Full Band	Required	MET

5. ANALYSIS

5.1 Radar Sub-System Analysis

The radar sub-system demonstrated superior performance across all specified test scenarios. The AESA array achieved a beam-forming accuracy of $\pm 0.03^\circ$ throughout the temperature range, significantly exceeding the $\pm 0.10^\circ$ specification. Signal processing latency remained below 6 ms for all target types evaluated, enabling real-time tracking of fast-maneuvring targets at ranges up to 700 km.

The enhanced ECCM suite, comprising wideband frequency agility, advanced pulse compression, and multi-channel adaptive sidelobe cancellation, was validated against next-generation jamming platforms provided by the Electronic Warfare (EW) Wing. The system successfully maintained track on 91% of targets even under heavy barrage jamming conditions at a jamming-to-signal ratio of 83 dB/MHz.

5.2 Data Link and Communications Analysis

The tactical data link demonstrated consistent performance at ranges up to 450 km, utilising the enhanced Link 16-compatible waveform developed indigenously by DRDO-DLRL, Hyderabad. Throughput of 2.24 Mbps was consistently achieved during ground and airborne test phases, sufficient to support 300 simultaneous track updates per second.

5.3 Reliability, Maintainability and Supportability

A total of 9 minor corrective maintenance actions were recorded over the 45-day trial period. The Mean Time to Repair (MTTR) averaged 1.8 hours, well within the 4-hour specification. The Built-In Test Equipment (BITE) detected 96% of fault conditions automatically, reducing dependence on manual fault isolation procedures.



Figure 2: Enhanced Satellite Data Link Configuration – ASRS Mk II Integration



Figure 3: System Performance Analysis – ASRS Mk II (Phase III Field Trials)

6. CONCLUSION

The ASRS Mk II system has successfully completed Phase III field trials with all Key Performance Indicators significantly exceeded. The system demonstrated a detection range of 750 km, track capacity of 300 simultaneous targets, and an MTBF of 650 hours – representing a transformative leap in indigenous airborne surveillance capability. The overall system availability of 98.1% confirms readiness for operational deployment with the Indian Air Force.

The enhanced AESA radar, next-generation signal processor, Extended Data Link, and advanced mission computer constitute a fully indigenous technology baseline. The ASRS Mk II represents a strategic milestone in DRDO's pursuit of world-class self-reliant defence electronics.

7. RECOMMENDATIONS

- Immediate clearance of ASRS Mk II for Limited Series Production of 5 platforms
- Initiation of ASRS Mk III programme with target detection range exceeding 1,000 km
- Establishment of a dedicated MRO facility at Air Force Station Lohegaon for ASRS platform support
- Integration of Advanced IFF Mode 5 and Mode 4 in all production configurations
- Commencement of joint trials with the Navy for maritime surveillance variant development
- Formation of a Technology Transfer Working Group with HAL for indigenous production scaling

DOCUMENT APPROVAL

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